**Process Analysis**

**Inputs → Transformation → Outputs**

| **Definitions** | |
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| ***Cycle Time*** | Average time between completions of successive units in a process (time/unit) |
| ***Throughput Time*** | Average time that it takes a unit to move through an entire process. Includes processing and blocking time (time) |
| ***Throughput Rate*** | Output rate that the process is expected to produce over a period of time (units/time) |
| *Utilization* | Ratio of time resource is used relative to that available (used/theoretical) |
| *Bottleneck* | A resource that limits the capacity of maximum output of the process. Can be when an employee works too slowly in a multi-stage process and their work piles up. |
| *Pacing* | Fixed timing of the movement of items through the process |
| *Blocking* | Occurs when activities in the stage must stop because there is no place to deposit the item just completed (**adds to throughput time**) |
| *Starving* | Occurs when the activities in a stage must stop because there is no work (**does not add to throughput time**) |
| *Buffer* | A storage area between stages where the output of a stage is placed prior to being used in a downstream stage (holding area). |
| *Efficiency* | A ratio of the actual output of a process relative to some standard |
| *Pull - Make to order* | System activated in response to an actual order. WIP and finished goods inventory is kept to a minimum |
| *Push- Make to stock* | Process or system activated to meet expected or forecast demand. Customer orders served from target stock levels |
| *Process Flowcharting* | Use of a diagram to present major elements of a process (tasks/operations, flow of materials/customers, decision point, storage areas/queues, etc). Often a first step in analyzing a process. Provides a good visual for those not familiar with the process. Helps identify waste! |
| *Methods to Improve Process Performance* | Perform activities in parallel, change the sequence of activities, reduce interruptions, etc… |

| **Equations** | |
|---|---|
| ***Little's Law*** *(assumes process is in steady state!)* | Book's definition: Inventory = Throughput rate X Flow Time<br>$Throughput\ (time) = \frac{WIP\ (units)}{throughput\ rate\ (units/time)}$<br>$Throughput\ (time)\ = WIP\ (units)\ *$ cycle time (time/unit) |
| *Throughput Rate and Cycle Time* | throughput rate = 1/cycle time → cycle time = 1/throughput rate |
| *Efficiency* | $Efficiency = \frac{\sum Processing\ Time}{N*CT}$ = Actual / Standard<br>N = number of stations CT = cycle time |
| Velocity | $\text{Velocity} = \frac{\text{value added time}}{\text{throughput time}}$ |